

GRAND MANAN, NB, Canada

WMO: 710300

Lat: 44.712N

Lon: 66.802W

Elev: 78

StdP: 100.39

Time Zone: -4.00 (NAA)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-15.5	-23.2	0.5	-16.3	-21.0	0.6	-13.6	12.7	0.5	11.0	0.0	4.7	300	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.6	26.2	19.5	24.6	18.5	23.1	17.8	21.0	24.6	19.9	22.9	18.9	21.5	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	22.5	18.7	13.7	21.3	17.9	13.0	20.2	61.1	24.8	57.2	22.9	54.0	21.6	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	7.8	DB	-21.4	29.7	2.4	1.5	-23.1	30.8	-24.5	31.7	-25.8	32.5	-27.5	33.6	(4)
(5)			WB	-21.6	23.1	2.4	1.1	-23.3	23.9	-24.7	24.5	-26.0	25.2	-27.8	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-4.9	-4.3	-0.8	4.5	9.4	14.0	17.6	17.5	14.0	8.7	3.9	-1.3	(6)	
	DBStd	8.82	5.76	5.08	4.52	3.27	3.05	3.08	2.48	2.43	3.53	3.73	4.27	5.16	(7)	
	HDD10.0	2035	462	401	335	168	48	4	0	0	7	72	187	351	(8)	
	HDD18.3	4339	721	634	593	416	278	134	43	43	135	300	433	609	(9)	
	CDD10.0	784	0	0	1	2	29	124	236	233	126	30	4	0	(10)	
	CDD18.3	47	0	0	0	0	0	4	20	18	4	0	0	0	(11)	
	CDH23.3	291	0	0	1	0	8	40	119	98	24	1	0	0	(12)	
	CDH26.7	37	0	0	0	0	1	4	16	12	4	0	0	0	(13)	
(14) Wind	WSAvg	3.7	4.3	4.4	4.4	3.9	3.6	3.2	2.8	2.7	3.0	3.6	3.9	4.3	(14)	
(15) Precipitation	PrecAvg	1272	115	93	113	94	101	93	81	81	101	126	123	134	(15)	
	PrecMax	1802	211	191	177	206	208	256	192	233	215	332	195	317	(16)	
	PrecMin	743	49	28	24	13	30	33	19	15	38	37	69	48	(17)	
	PrecStd	211	37	35	40	49	42	54	48	47	50	68	37	61	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	9.0	13.3	17.0	24.3	27.2	28.8	28.5	26.0	21.4	14.9	11.9	(19)	
		MCWB	10.1	7.9	7.7	10.6	16.0	19.5	21.3	21.6	20.0	15.4	13.5	11.5	(20)	
	2%	DB	8.2	6.5	8.9	13.6	19.9	24.1	26.1	25.6	23.2	17.6	13.6	10.2	(21)	
		MCWB	7.7	5.3	5.7	8.4	13.6	17.5	19.7	19.1	17.9	14.6	12.7	9.8	(22)	
	5%	DB	6.0	4.7	6.8	11.5	17.2	21.9	24.5	24.2	21.0	15.9	12.4	8.4	(23)	
		MCWB	5.1	3.6	5.0	7.5	11.8	16.0	18.7	18.4	16.8	14.0	11.4	7.6	(24)	
	10%	DB	3.6	3.2	5.2	9.8	14.8	20.0	22.9	22.7	19.4	14.5	10.9	6.6	(25)	
		MCWB	2.6	2.1	3.3	6.8	10.6	14.8	17.9	17.9	16.0	12.9	9.8	5.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	8.5	9.6	12.1	17.9	20.4	22.4	22.5	21.0	17.5	14.4	11.6	(27)	
		MCDB	10.4	8.8	11.8	14.6	22.8	25.4	26.5	26.9	24.5	18.9	14.6	11.7	(28)	
	2%	WB	7.7	5.8	7.0	10.2	14.6	18.9	21.0	21.0	19.1	15.7	13.1	9.8	(29)	
		MCDB	8.0	6.2	7.9	11.5	18.3	22.9	24.8	24.2	21.6	16.8	13.3	10.1	(30)	
	5%	WB	5.3	3.9	5.3	8.7	12.9	17.2	19.9	19.8	18.0	14.5	11.7	8.0	(31)	
		MCDB	5.9	4.5	6.5	10.5	15.7	20.4	23.0	22.4	19.8	15.4	12.1	8.3	(32)	
	10%	WB	2.7	2.4	3.6	7.3	11.5	15.8	18.9	18.9	17.0	13.2	10.0	5.7	(33)	
		MCDB	3.6	3.2	4.9	9.1	13.8	18.6	21.7	20.9	18.4	14.3	10.6	6.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.5	7.8	8.0	9.0	9.4	9.4	9.6	9.7	8.5	7.6	7.6	(35)	
		MCDBR	10.3	9.3	9.4	11.3	13.5	13.1	12.1	11.9	11.0	9.8	8.2	9.4	(36)	
	5% WB	MCWBR	10.7	9.0	7.5	7.3	7.9	7.4	6.7	6.7	6.9	7.3	7.7	9.8	(37)	
		MCWBR	10.3	9.3	8.5	8.7	10.7	11.1	10.2	9.4	8.3	7.9	7.5	9.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.298	0.301	0.328	0.353	0.372	0.407	0.424	0.400	0.364	0.335	0.317	0.300	(40)	
		taud	2.477	2.493	2.454	2.425	2.418	2.343	2.323	2.391	2.455	2.530	2.534	2.491	(41)	
	Ebn at Noon		827	895	909	910	898	863	843	852	858	842	799	785	(42)	
		Edh at Noon	69	81	97	109	113	122	124	112	97	78	66	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.41	2.31	3.55	4.46	5.05	5.54	5.65	5.04	4.12	2.65	1.54	1.12	(44)	
	RadStd		0.12	0.19	0.26	0.43	0.61	0.50	0.44	0.32	0.28	0.26	0.15	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.43	N/A	N/A	+54	N/A

Nomenclature: See separate page